SIMATS ENGINEERING

ASSESSMENT TOOL 3: Debugging

COURSE NAME:Query Processing for Data Science

COURSE CODE: DSA0503

COURSE OUTCOME COVERED

CO2: Use Python basics and SQL libraries to connect to databases SQLite, MySQL, PostgreSQL and perform CRUD operations like Create, Read, Update, Delete. (BTL6)

Assessment Tool Weightage: 25%

Question Count: 10 **Total Marks: 20**

Code of Conduct

I VISHWA T/ 192424180 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

VISHWA.T

Debugging

Criteria	Problem Identification (4 Marks)	Debugging (4 Marks)	Optimization (4 Marks)	Analysis (4 Marks)	Code Quality (4 Marks)
Marks					

Total Marks Awarded:

Questions:

Question 1: SQLite Database Connection

The following Python program is written to connect to an SQLite database and create a table. However, the program raises an error and does not execute successfully. Analyze the code, identify the mistakes, and rewrite the corrected version.

```
import sqlite3

conn = sqlite3.connect("student.db")

cursor = conn.cursor()

cursor.execute("""

CREATE TABLE IF NOT EXISTS Student(

    Student_ID INTEGER PRIMARY KEY,

    Name TEXT,

    Department TEXT

)

""")

conn.commit()

conn.close()

print("Student table created successfully")
```

Output:

```
Student table created successfully
```

Question 2: MySQL Record Insertion

The following Python program is intended to connect to a MySQL database and insert a new student record. The program contains multiple errors that prevent successful execution. Debug the code and provide the corrected program.

```
import mysql.connector

conn = mysql.connector.connect(

    host="localhost",

    user="root",

    password="root",
```

```
    database="college"
)
cursor = conn.cursor()
query = """
INSERT INTO Student(Student_ID, Name, Department)
VALUES (%s, %s, %s)
"""
cursor.execute(query, (101, "Alice", "CSE"))
conn.commit()
print("Record inserted successfully")
cursor.close()
conn.close()
```

Output:

```
Record inserted successfully
```

Question 3: PostgreSQL Data Retrieval

The following Python program is designed to retrieve all records from the Employee table in a PostgreSQL database. The code contains logical and syntax errors. Analyze the program, identify the errors, and write the corrected version.

```
import psycopg2
conn = psycopg2.connect(
    host="localhost",
    database="company",
    user="postgres",
    password="12345678"
)
cursor = conn.cursor()


cursor.execute("SELECT * FROM Employee")
```

rows = cursor.fetchall()

for row in rows:

    print(row)

cursor.close()

conn.close()

Output:

```
(101, 'John', Decimal('55000.00'))
(102, 'David', Decimal('60000.00'))
(103, 'Alice', Decimal('65000.00'))
```

Question 4: SQLite Record Update

The following program attempts to update the department of a student in an SQLite database. The update operation fails due to errors in the code. Identify the mistakes and rewrite the corrected program.

import sqlite3

conn = sqlite3.connect("college.db")

cursor = conn.cursor()

cursor.execute("""

CREATE TABLE IF NOT EXISTS Student(

    Student_ID INTEGER PRIMARY KEY,

    Name TEXT,

    Department TEXT

)

""")

cursor.execute("""

INSERT OR IGNORE INTO Student

VALUES (105, 'John', 'CSE')

""")

```
cursor.execute("""
UPDATE Student
SET Department = ?
WHERE Student_ID = ?
""", ("ECE", 105))
conn.commit()
print("Record updated successfully")
cursor.close()
conn.close()
```

Output:

```
Record updated successfully
```

Question 5: PostgreSQL Record Deletion

The following Python program is intended to delete a customer record from a PostgreSQL database. The code contains errors related to SQL execution and resource management. Debug the program and provide the corrected version.

```
import psycopg2
conn = psycopg2.connect(
    host="localhost",
    database="bank",
    user="postgres",
    password="YOUR_POSTGRES_PASSWORD"
)
cursor = conn.cursor()
customer_id = 105
query = """
DELETE FROM Customer
WHERE Customer_ID = %s
```

```
"""

cursor.execute(query, (customer_id,))

conn.commit()

print("Record deleted successfully")

cursor.close()

conn.close()
```

Output:

```
Record deleted successfully
```

Question 6: SQLite Record Retrieval

The following Python program is intended to retrieve the details of a student whose Student_ID is 105 from an SQLite database. The program produces an error and does not display the required record. Analyze the code, identify the mistakes, and rewrite the corrected version.

```
import sqlite3
conn = sqlite3.connect("college.db")
cursor = conn.cursor()
cursor.execute("""
CREATE TABLE IF NOT EXISTS Student(
    Student_ID INTEGER PRIMARY KEY,
    Name TEXT,
    Department TEXT
)
""")

cursor.execute("""
INSERT OR IGNORE INTO Student
VALUES (105, 'John', 'CSE')
""")
student_id = 105
query = """
SELECT * FROM Student
WHERE Student_ID = ?
"""
cursor.execute(query, (student_id,))
row = cursor.fetchone()
if row:
    print("Student Record:")
    print(row)
else:
    print("Student record not found")

cursor.close()
conn.close()
```

Output:

```
Student Record:
(105, 'John', 'ECE')
```

Question 7: MySQL Table Creation

A student wrote the following Python program to create a table in a MySQL database. The program contains syntax and logical errors. Debug the code and provide the corrected version.

```
import mysql.connector

conn = mysql.connector.connect(

    host="localhost",

    user="root",

    password="YOUR_MYSQL_PASSWORD",

    database="college"

)

cursor = conn.cursor()

cursor.execute("""

CREATE TABLE IF NOT EXISTS Student(

    Student_ID INT PRIMARY KEY,

    Name VARCHAR(50),

    Department VARCHAR(30)

)

""")

conn.commit()

print("Student table created successfully")

cursor.close()

conn.close()
```

Output:

```
Student table created successfully
```

Question 8: PostgreSQL Record Insertion

The following Python program is intended to insert a new employee record into a PostgreSQL database. The program fails during execution due to multiple errors. Analyze the code, identify the mistakes, and rewrite the corrected version.

```
import psycopg2
conn = psycopg2.connect(
    host="localhost",
    database="company",
    user="postgres",
    password="YOUR_POSTGRES_PASSWORD"
)
cursor = conn.cursor()
sql = """
INSERT INTO Employee(Employee_ID, Name, Salary)
VALUES (%s, %s, %s)"""
cursor.execute(sql, (104, "John", 55000))
conn.commit()
print("Employee record inserted successfully")
cursor.close()
conn.close()
```

Output:

```
Employee record inserted successfully
```

Question 9: MySQL Record Update

The following Python program is intended to update the salary of an employee in a MySQL database. The update operation does not execute successfully. Debug the code and provide the corrected version.

```
import mysql.connector

conn = mysql.connector.connect(

    host="localhost",

    user="root",

    password="YOUR_MYSQL_PASSWORD",

    database="company"

)

cursor = conn.cursor()

salary = 65000

emp_id = 110
```

```
query = """

UPDATE Employee

SET Salary=%s

WHERE Employee_ID=%s

"""

cursor.execute(query, (salary, emp_id))

conn.commit()

print("Record Updated")

cursor.close()

conn.close()
```

Output:

```
Record Updated
```

```
mysql> SELECT * FROM Employee WHERE Employee_ID=110;
+-------------+-------+----------+
| Employee_ID | Name  | Salary   |
+-------------+-------+----------+
|         110 | David | 65000.00 |
+-------------+-------+----------+
1 row in set (0.00 sec)
```

Question 10: PostgreSQL Record Deletion

The following Python program attempts to delete all customer records belonging to the city Chennai. However, the program contains errors related to parameter passing and transaction handling. Analyze the code, identify the mistakes, and rewrite the corrected version.

```
import psycopg2

conn = psycopg2.connect(

    host="localhost",

    database="bank",
```

```
    user="postgres",
    password="YOUR_POSTGRES_PASSWORD"
)
cursor = conn.cursor()
city = "Chennai"
query = """
DELETE FROM Customer
WHERE City = %s
"""
cursor.execute(query, (city,))
conn.commit()
print("Records Deleted")
cursor.close()
conn.close()
```

Output:

```
Records Deleted
```

	customer_id [PK] integer	name character varying (50)	city character varying (50)
1	202	Kumar	Bangalore

Signature of the Course Faculty